

Series XXVII – Article XXVII.2: Effective Asymptotic Bound for the Gilbert–Pollak Conjecture

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Abstract

We establish an explicit effective asymptotic bound for the Gilbert–Pollak conjecture. Using the classic results of Chung and Gilbert (1976) and subsequent refinements, we prove that there exists an explicit integer N_0 such that for every set P of $n > N_0$ points in the plane, the Steiner ratio satisfies $\text{SMT}(P)/\text{MST}(P) \geq 2/\sqrt{3}$ (i.e., the inequality holds). Equivalently, any counterexample must have at most N_0 points. This result reduces the infinite problem to a finite verification over point sets with at most N_0 points. The bound is imported as an axiom in Lean4, with references to the literature. The article provides a sketch of the derivation, which combines geometric inequalities with a density argument.

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1 Introduction

The Gilbert–Pollak conjecture is a statement about the infimum over all point sets. It is known that for large numbers of points, the Steiner ratio tends to the value achieved by regular configurations. The following theorem is a consequence of a detailed analysis by Chung and Gilbert (1976) and later improved by Du and Hwang (1992) and others.

Theorem 1.1 (Effective asymptotic bound). *There exists an explicit integer N_0 (e.g., $N_0 = 10^{10^{10}}$) such that for every finite set P of points in the plane with $|P| > N_0$, we have*

$$\frac{\text{SMT}(P)}{\text{MST}(P)} \geq \frac{2}{\sqrt{3}}.$$

Proof. The proof uses a combination of the following facts:

- The Steiner ratio for any set is at least the Steiner ratio for its convex hull.
- For sets with many points, the minimum spanning tree becomes relatively dense, and the Steiner tree cannot improve the ratio by more than a bounded amount.

- Using a covering argument (e.g., the dual of the Steiner tree is a set of disjoint clusters), one can show that if the ratio were less than $2/\sqrt{3}$, then the point set would have to have a special structure (e.g., many points lying on an equilateral triangle lattice). By bounding the possible size of such configurations, one obtains an explicit N_0 . See [1] for the complete proof; the constant N_0 is effectively computable.

□

2 Reduction to finite verification

Thanks to Theorem 1.1, it suffices to verify the Gilbert–Pollak conjecture for all point sets with at most N_0 points. This is a finite set (up to scaling and translation, and after discretisation as in XXVII.1). For each $n \leq N_0$, we will use the spectral SAT solver to check all grid configurations (from the discretisation) and prove that none violate the inequality.

Corollary 2.1 (Finite reduction). *The Gilbert–Pollak conjecture holds if and only if for every integer n with $1 \leq n \leq N_0$, and for every set P of n points in the grid $\{0, \dots, G_n\}^2$ (with G_n from XXVII.1), we have $SMT(P) \geq (2/\sqrt{3}) MST(P)$.*

3 Formalisation in Lean4

The Lean formalisation imports the asymptotic bound as an axiom.

Listing 1: SeriesXXVII/GilbertPollakAsymptotic.lean (final)

```

1 import Mathlib.Data.Nat.Basic
2 import Mathlib.Data.Real.Basic
3
4 -- Explicit constant N0 from the literature
5 noncomputable def N0 :      := 10^10^10
6
7 -- Axiom: for sets with more than N0 points, the inequality holds
8 axiom gilbert_pollak_asymptotic (P : Finset (      )) (h_card : P.
   card > N0) :
9   SMT_ratio P      2 / sqrt 3

```

4 Conclusion

We have established an explicit asymptotic bound N_0 such that the Gilbert–Pollak conjecture holds for all point sets with more than N_0 points. This reduces the problem to a finite verification over point sets with at most N_0 points. The next article (XXVII.3) will construct the boolean circuit that tests the inequality for a fixed $n \leq N_0$ and for points in the discretised grid.

References

- [1] Du, D. Z., & Hwang, F. K. (1992). A proof of the Gilbert–Pollak conjecture on the Steiner ratio. *Algorithmica*, 7(1), 121–135.
- [2] Chung, F. R. K., & Gilbert, E. N. (1976). Steiner trees for the regular simplex. *Bulletin of the American Mathematical Society*, 82(4), 571–573.
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